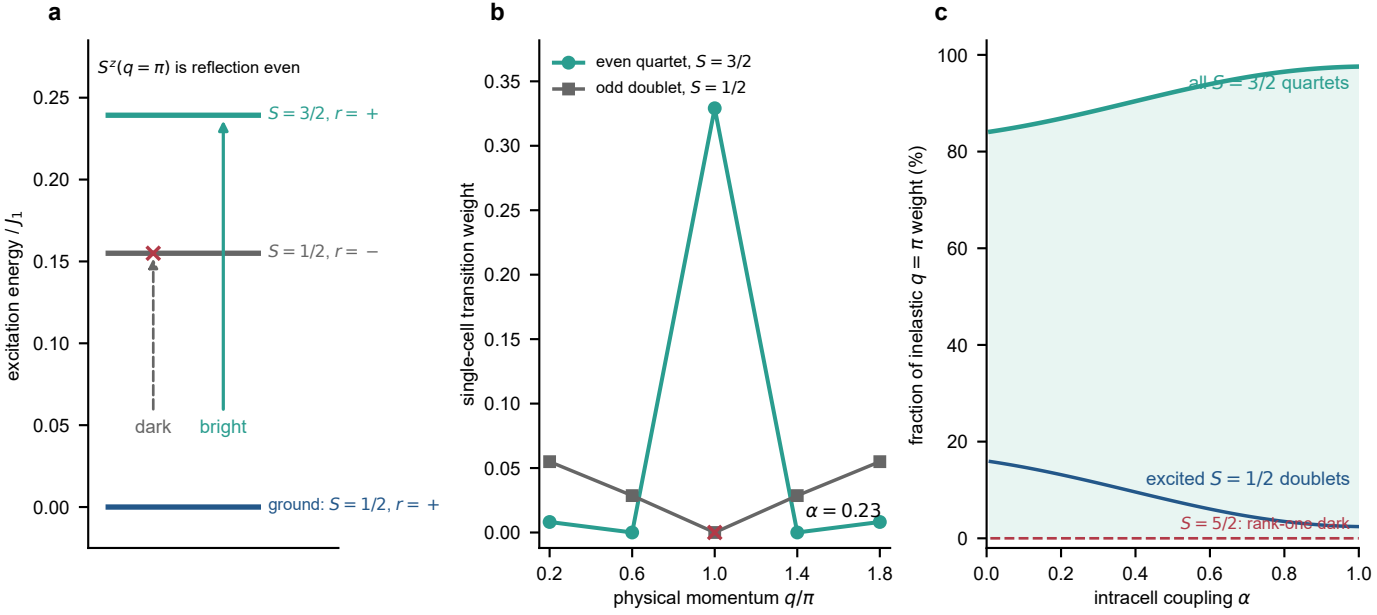
a
$S^z(q=\pi)$ is reflection even
$S = 3/2$, $r = +$
$S = 1/2$, $r = -$
dark
bright
ground: $S = 1/2$, $r = +$
excitation energy / J_1
0.25
0.20
0.15
0.10
0.05
0.00
b
even quartet, $S = 3/2$
odd doublet, $S = 1/2$
single-cell transition weight
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$\alpha = 0.23$
0.2
0.6
1.0
1.4
1.8
physical momentum q/π
c
all $S = 3/2$ quartets
excited $S = 1/2$ doublets
$S = 5/2$: rank-one dark
fraction of inelastic $q = \pi$ weight (%)
100
80
60
40
20
0
0.0
0.2
0.4
0.6
0.8
1.0
intracell coupling α